



There has been a definite change towards digital stations as these help in efficient working temperature, which is needed when working with temperature-sensitive components—Manish Kwatra, CEO, Metro Electronic



MetroQ MTQ 915

shooting the set temperature. Additionally, all conventional systems require calibration.

SmartHeat technology overcomes this problem by offering variable power with fixed temperature. It senses the exact thermal requirement for each solder joint and responds by delivering the precise amount of thermal energy required to create a reliable connection. The technology allows users to operate at lower soldering temperatures and eliminates the risk of temperature overshoot.

Where to look

Sumitron offers soldering and desoldering solutions from Hakko Corp., Japan.

Goot is another Japanese company whose products are sold in India by Advance Tech Services.

OK International offers soldering and desoldering solutions under its Metcal and Oki brands.

Metro Electronic caters to three categories: PTH (with brand MetroQ), SMD (with brand MAX GOLD) and irons (with brand Soldron). The MetroQ brand is owned by Metro Electronic, while it's the distributor for

MAX GOLD and Soldron brands.

Blue Sky offers both ESD-safe (BS-936) and lead-free (BS-942) soldering stations.

Hexaa Bytes Systems provides soldering and desoldering stations from indigenous as well as international manufacturers for PTH and SMD components.

Emerging areas of opportunities

SMT is now fairly well entrenched within the electronics segment in India. The compliance to ROHS directive is also growing. With the increased learning curve and the booming domestic market, many

is also looking up.

Latest trends in soldering are:

1. Increased demand for no-clean flux: An ideal flux must bind effectively to make sound solder joints and leave minimal amount of residue after soldering. The no-clean flux meets both these requirements.

2. Selective soldering (using nozzles): It is increasingly being preferred to traditional wave- (using pallets) and hand-soldering techniques due to the enhanced capabilities in terms of cost, defect reduction, solder quality and improved throughput. Lead-free soldering has enabled selective soldering technology to gain more prominence and wider adoption. Industrial, telecommunication, aerospace and medical are some of the areas where opportunities abound for selective soldering. The technology is appropriate for electronics assembly manufacturers dealing with mixed-technology boards.



The latest soldering stations have their tip integrated with the heating element, allowing constant temperature and quick heat generation and recovery—Ajay Gupta, director-technical, Advance Tech Services

companies are looking at India as a possible manufacturing destination. The Indian electronics manufacturing market has already been growing by over 15 per cent for the last five years. Most of the industry reports indicate that the growth would continue. This, in turn, would bring the market growth for electronics assembly products in India.

The emerging areas of opportunities for soldering and desoldering stations are telecommunication, aerospace and medical. Solar energy sector

Advice for Indian manufacturers

There has been very little technological advancement in the products made by Indian manufacturers. We should invest more in R&D so as to become on par with foreign counterparts. Focus on quality, not the price, is needed. At present, most manufacturers look at reducing the price while doing the development. They forget the end-user goal and facility.

"To overcome the gap in quality, Indian manufacturers need to switch to fully automated manufacturing process, which is faster as it hardly requires manual effort," says Rajput. ■



To overcome the gap in quality, Indian manufacturers need to switch to fully automated manufacturing process—Mukesh Rajput, CTO, Max Technology & Co.

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